

A Model of Bank Failures: Funding Frictions and the Dynamics Before Collapse

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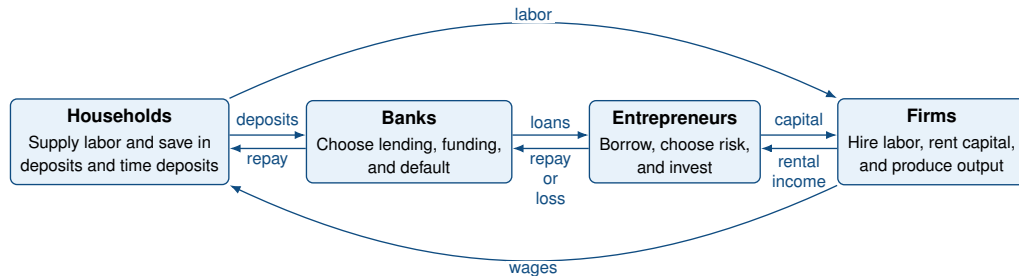
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What this paper does

- ▶ **Empirical motivation:**
 - ▶ Before failure, banks increase leverage and rely more on time deposits.
 - ▶ Profitability declines while loan losses rise closer to failure.
- ▶ **Setup:** model of banks with risky capital (loans) and two choices of debt, s.t. deposits (savings+ checking) and l.t. time deposits (certificate of deposits)
- ▶ **Question:** How do funding and debt dilution shape the buildup of bank distress and eventual failure?
- ▶ **Mechanism:**
 - ▶ Time deposits reduce rollover risk and the need for inefficient asset liquidation.
 - ▶ Debt dilution encourages additional debt issuance as banks weaken.
- ▶ **Result:** Liability-side incentives help explain the gradual balance-sheet deterioration preceding bank failure.

Model environment



- ▶ Banks face liquidity shocks, limited commitment, and self-fulfilling runs.
- ▶ Banks cannot issue equity, and reducing lending entails liquidation costs.
- ▶ Higher lending rates induce entrepreneurs to choose riskier capital projects.
- ▶ Bank default occurs when continuation becomes infeasible.

Bank problem: the relevant tradeoff

$$V^c(z, n, \ell, b) = \max_{R'_\ell, d', b'} \left\{ \underbrace{\text{div}}_{\text{today}} + \underbrace{\mathbb{E}[\sigma \Lambda h(z', n', \ell', b', \xi') V^c(z', n', \ell', b')]}_{\text{future}} \right\}$$

Bank choices

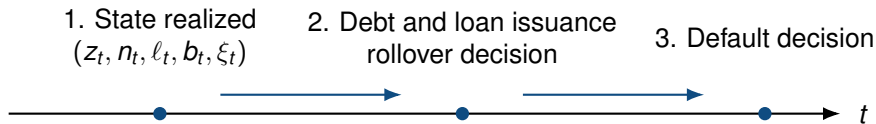
- ▶ Lending rate
set their rate taking $\ell^d(R'_\ell, z, r^e)$
- ▶ Deposits
mature every period
- ▶ Time deposits
pay coupon, fraction λ mat. every period

Key frictions

- ▶ Non-negative dividends
- ▶ Positive expected equity
- ▶ Endogenous debt prices
- ▶ Self-fulfilling runs
- ▶ Costly liquidation of assets
- ▶ Issuance and buyback costs for deposits

Funding choices affect current payouts but also shape future borrowing costs and default risk.

Timing creates rollover risk



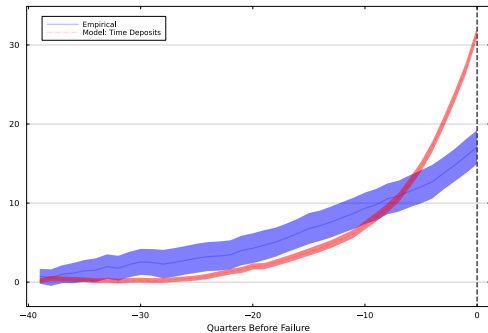
- ▶ Creditors price new funding before the default decision is realized.
- ▶ Pessimistic rollover beliefs can lower debt prices and validate default.

Model validation

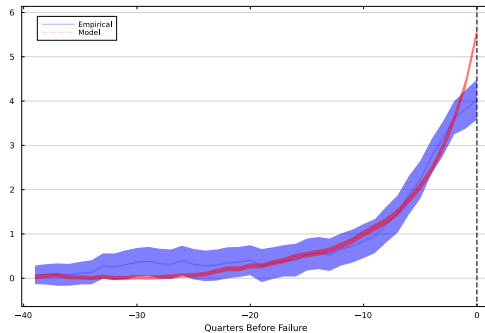
- ▶ Parameters are calibrated to match steady-state moments.
- ▶ I evaluate whether the model reproduces the **untargeted dynamics** preceding bank failures.
- ▶ The comparison focuses on:
 - ▶ funding composition,
 - ▶ leverage,
 - ▶ profitability and loan losses (appendix).

The transition toward failure is not targeted in the calibration.

Model validation



Time deposits



Leverage

Counterfactuals

No time deposits

- ▶ Eliminates long-term funding.
- ▶ Increases rollover risk.
- ▶ Isolates the role of funding maturity.

No debt dilution

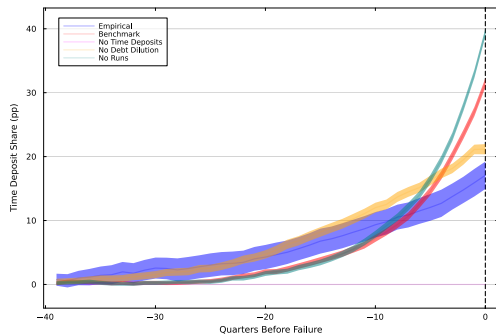
- ▶ Covenant protects existing creditors.
- ▶ Eliminates debt dilution.
- ▶ Isolates leveraging incentives.

No runs

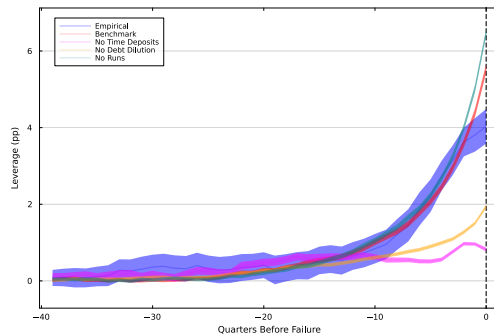
- ▶ Eliminates self-fulfilling runs.
- ▶ Removes refinancing crises.
- ▶ Isolates leverage incentives.

Which mechanism explains the pre-failure dynamics?

Counterfactual: debt dilution



Time deposits



Leverage

- Eliminating debt dilution reduces leverage and, consequently, banks' reliance on time deposits despite their higher market value.

Aggregate implications

- ▶ Eliminating debt dilution generates the largest stability gain: failures fall by about 23 percent relative to the benchmark.
- ▶ Eliminating self-fulfilling runs has a smaller effect on aggregate outcomes.
- ▶ Eliminating time deposits increases failures by roughly fourfold, highlighting their stabilizing role.
- ▶ Aggregate capital, output, consumption, and welfare move little across counterfactuals.

Long-term funding stabilizes banks without substantially changing aggregate lending.

Conclusion

- ▶ Banks approaching failure gradually shift toward time deposits, increase leverage, and experience worsening profitability and loan performance.
- ▶ Long-term funding has a dual role: it reduces rollover risk but creates debt-dilution incentives.
- ▶ Debt dilution primarily distorts liability choices; self-fulfilling runs and asset liquidation primarily affect rollover risk.
- ▶ **Takeaway:** liability-side incentives are important to understand the path from distress to failure.

Appendix

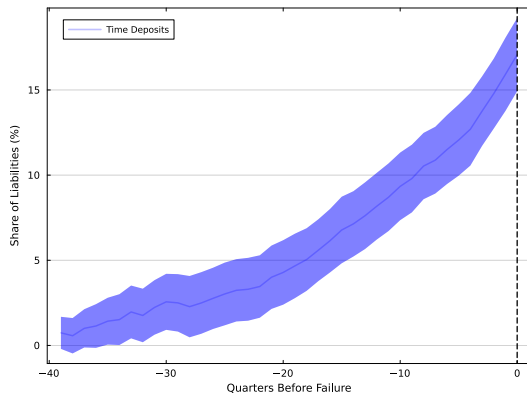
Backup: event-study specification

$$y_{j,t} = \alpha_j + \alpha_t + \sum_{k=-40}^0 \mathbb{I}_{k=t} \beta_k + \varepsilon_{j,t}.$$

- ▶ k indexes quarters to failure.
- ▶ The omitted period is $k = -40$, ten years before failure.
- ▶ Coefficients measure the evolution of failing banks relative to banks that never failed in the sample.

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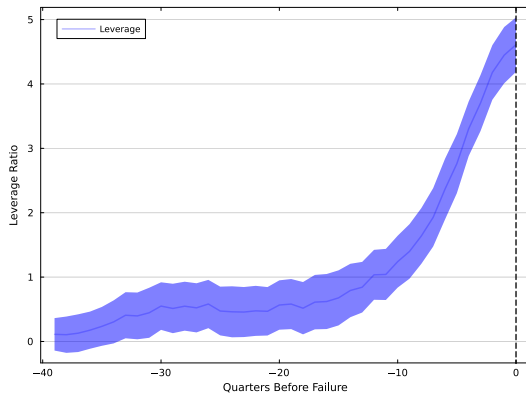
Empirical motivation: banks rely more on long-term funding



- ▶ Time deposits increase steadily as banks approach failure.
- ▶ Banks substitute toward longer-maturity funding.

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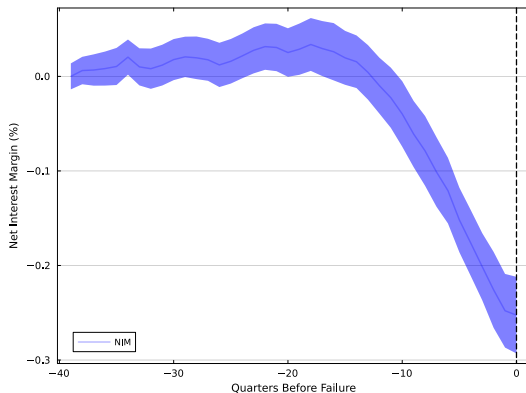
Empirical motivation: banks increase leverage



- Leverage rises persistently before default.

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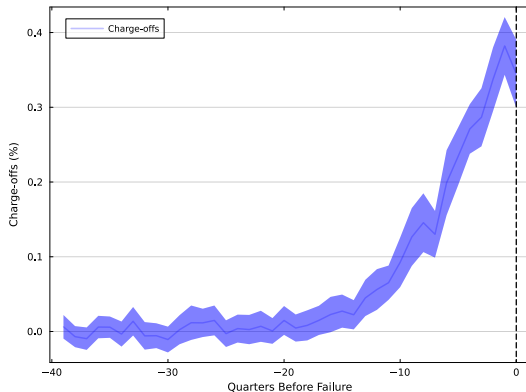
Empirical motivation: profitability deteriorates



- ▶ Net interest margins compress before failure.
- ▶ Lower profitability slows the rebuilding of bank net worth.

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Empirical motivation: credit losses increase



- ▶ Loan losses rise sharply as banks approach failure.
- ▶ Deteriorating asset quality precedes default by several quarters.

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Table 1: Calibrated Parameters

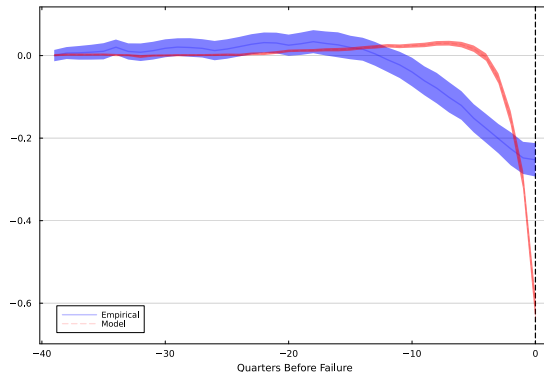
Parameter	Description	Calibrated Value
σ_z	Std. dev. of credit-quality shock	0.9
σ_ω	Std. dev. of i.i.d. shock	0.0175
ϕ	Fixed operating cost	10^{-3}
ζ	Deposit issuance cost	9×10^{-4}
θ	Shape of loan demand	1.9
σ	Manager myopia	0.98
ψ_0	Success-probability intercept	1.645
ψ_1	Success-probability sensitivity to risk	324
η	Conditional run probability	0.01
γ	Recovery rate on assets	0.9
r^e	Steady-state entrepreneurial return	0.017
δ^*	Depreciation rate	0.0143
$\bar{n}$	Equity of new banks	0.02
$\bar{k}$	Loan amount	98.95
ϑ	Scale of the Loan demand	0.011

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Table 2: Targeted Moments: Data vs. Model

Moment	Description	Data (%)	Model (%)
<i>Means</i>			
$\mathbb{E}[r^e x]$	Entrepreneurial return	2.75	2.75
$\mathbb{E}[1 - h]$	Share of failing banks	0.10	0.10
$\mathbb{E}\left[\frac{d+b}{\ell}\right]$	Leverage	89.2	89.3
$\mathbb{E}\left[\frac{b}{b+d}\right]$	Share of time deposits	36.8	33.0
$\mathbb{E}[pr_{\ell} - ytm_{\text{debt}}]$	Net interest margin	1.25	1.21
$\mathbb{E}[(1 - p)\nu]$	Net charge-offs	0.19	0.20
Δ	Depreciation rate	2.5	2.5
<i>Standard deviations</i>			
$\sigma[(1 - p)\nu]$	Net charge-offs	0.32	0.20
$\sigma\left[\frac{d+b}{\ell}\right]$	Leverage	3.6	1.3
$\sigma\left[\frac{b}{b+d}\right]$	Share of time deposits	16.8	8.1
$\sigma[ytm^b - ytm^d]$	Spread between time and demand deposits	0.21	0.18
$\sigma[pr_{\ell} - ytm_{\text{debt}}]$	Net interest margin	0.24	0.10
<i>Elasticity</i>			
$\frac{\text{cov}(400 * R_{\ell}, \log(\ell))}{\text{var}(400 * R_{\ell})}$	Loan Elasticity	-50	-36

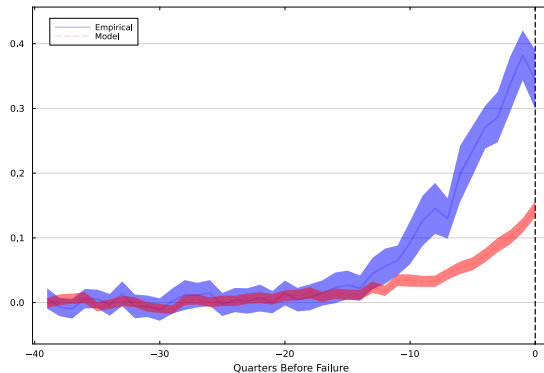
Model validation: profitability



- The model reproduces the decline in net interest margins before failure

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Model fit: loan losses



- Deteriorating asset quality contributes to the gradual erosion of bank net worth.

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Backup: resource constraint

$$\begin{aligned}div_t = & n_t + \tilde{q}_d(z, R'_\ell, d', b')d_{t+1} + \tilde{q}_b(z, R'_\ell, d', b')(b_{t+1} - (1 - \lambda)b_t) \\& - \ell^d(R_{\ell,t+1}, z_t, r_{t+1}^e) - Z(d_{t+1}, b_{t+1}, b_t) \\& - \phi - (1 - \gamma) \left(\frac{\max\{\ell_t - \ell^d(R_{\ell,t+1}, z_t, r_{t+1}^e)\}}{\ell_t} \right)^2 \ell_t \\& - \mathbf{1}_{\{b' < (1-\lambda)b\}} \left(q_b^{rf} - \tilde{q}_b(z, R'_\ell, d', b') \right).\end{aligned}$$

- ▶ Deposits fully mature every period.
- ▶ Only a fraction λ of time deposits matures each period.
- ▶ New issuance can dilute outstanding long-term creditors.

▶ Back to bank problem

Backup: default thresholds

- ▶ $n^f(z, \ell, b)$: lowest cash-on-hand for continuation under normal funding.
- ▶ $n^c(z, \ell, b)$: lowest cash-on-hand for continuation during a rollover crisis.

$$n^f(z, \ell, b) \leq n^c(z, \ell, b).$$

- ▶ If $n < n^f$: fundamental default.
- ▶ If $n^f \leq n < n^c$: bank-run state.
- ▶ The sunspot selects default in the run region.

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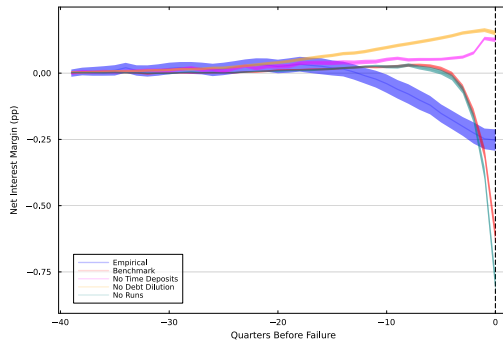
Backup: debt covenant counterfactual

$$CP(z, R'_\ell, d', b') = b(1 - \lambda) \left[q_b^{rf} - q_b(z, R'_\ell, d', b') \right].$$

- ▶ Banks compensate existing time-deposit holders for the decline in the value of outstanding claims.
- ▶ This eliminates debt dilution by construction.

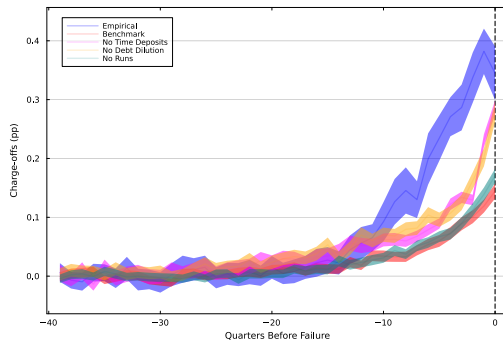
▶ Back to counterfactual

Counterfactual: profitability



- As banks become more levered and rely more on time deposits, higher funding costs largely offset the increase in lending rates, compressing net interest margins.

Counterfactual: loan losses



- ▶ Eliminating debt dilution shifts failures from run-driven to fundamental defaults.
- ▶ As a result, banks experience larger loan losses by the time they default.